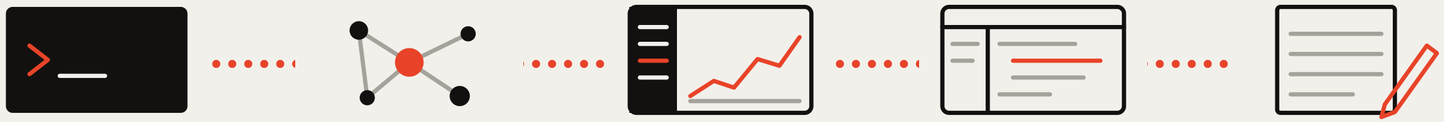
KaxaNuk
Sharing knowledge

INVESTMENT BOOTCAMP

CREATE YOUR RESEARCHER AND LEARN TO USE THE LAB.



01

Install a research companion, train it with papers, meet the Investment Lab, and see how a strategy is built, step by step, with your researcher beside you.

IN THIS SESSION

01 INSTALL IT

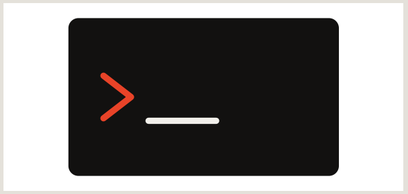
02 TRAIN IT

03 THE LAB

04 IN CODE

05 WRAP-UP

FROM AN EMPTY WINDOW TO YOUR FIRST STRATEGY'S MAP.

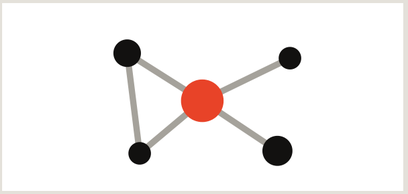


1 INSTALL

From GitHub, in three steps.

- Open your assistant, paste one line
- Tell it who you are; name it
- Its home, and your first read

You leave with: your researcher, named

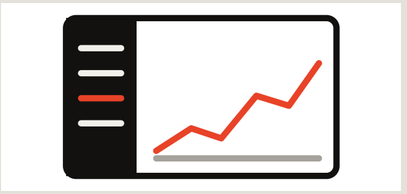


2 TRAIN

Papers in, skills to use them.

- Seventy years of ideas; one read
- Ask it, study, learn with it
- Watch it grow in Obsidian

You leave with: your first cited note



3 LAB

Its tools, and simple ideas to test.

- Six libraries, two ways in
- Simple ideas, ready to test
- Your researcher drives the Lab

You leave with: an idea to test



4 CODE

Where each part of a strategy lives.

- init-example or init-strategy
- Golden-Flow, from objective to attribution
- A copy on GitHub

You leave with: a strategy's map



5 WRAP-UP

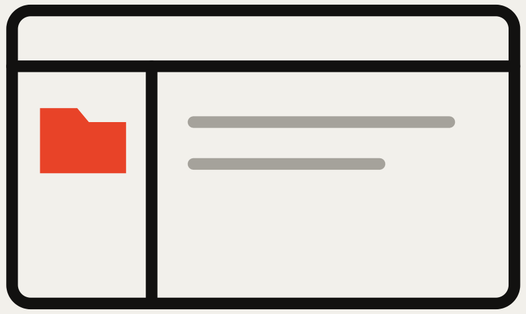
What you learned, and what comes next.

- The session's ideas, in your words
- Homework: use it on every project
- Next: the objective, in S02

You leave with: your plan for S02

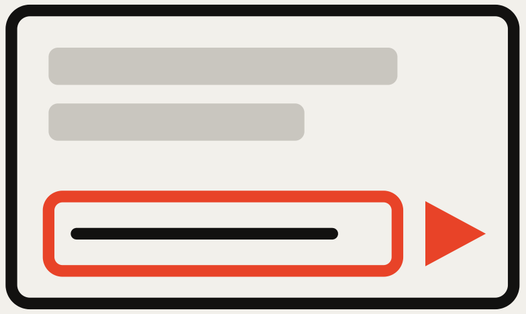
THREE STEPS TO CREATE YOUR RESEARCHER.

Build your research companion for all your projects, personal or work: test your ideas, build your investment strategies.



1 Open an assistant

An empty folder.



2 Paste one line

The README's line.

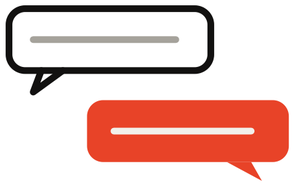


3 Tell it about you

Who you are, what you do, its name.

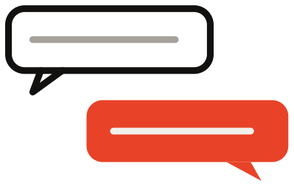
OPEN SOURCE ON GITHUB github.com/KaxaNuk/KaxaNuk-Researcher → Install — three steps, no coding

OPEN ONE ASSISTANT, SIGNED IN.



CLAUDE CODE

the Claude desktop app's Code tab, or the CLI



CODEX

the Codex app, or the CLI



GEMINI CLI

the CLI



CURSOR

the editor



COPILOT CLI

the CLI



A WEB CHAT

Cannot run commands.

Any one

Same researcher, same files.

What it reads

goes to your assistant's company, never to KaxaNuk.

COPY ONE LINE. SEND IT.

YOUR ASSISTANT · AN EMPTY FOLDER

Your language, from then on.

git and uv, if missing.

Allow this command? 2

A name: three ideas, or yours. 3

Two questions, about three minutes.

```
Please install this repo for me and set up my researcher, start to finish in
this conversation — install everything it needs without asking first:
https://github.com/KaxaNuk/KaxaNuk-Researcher
```

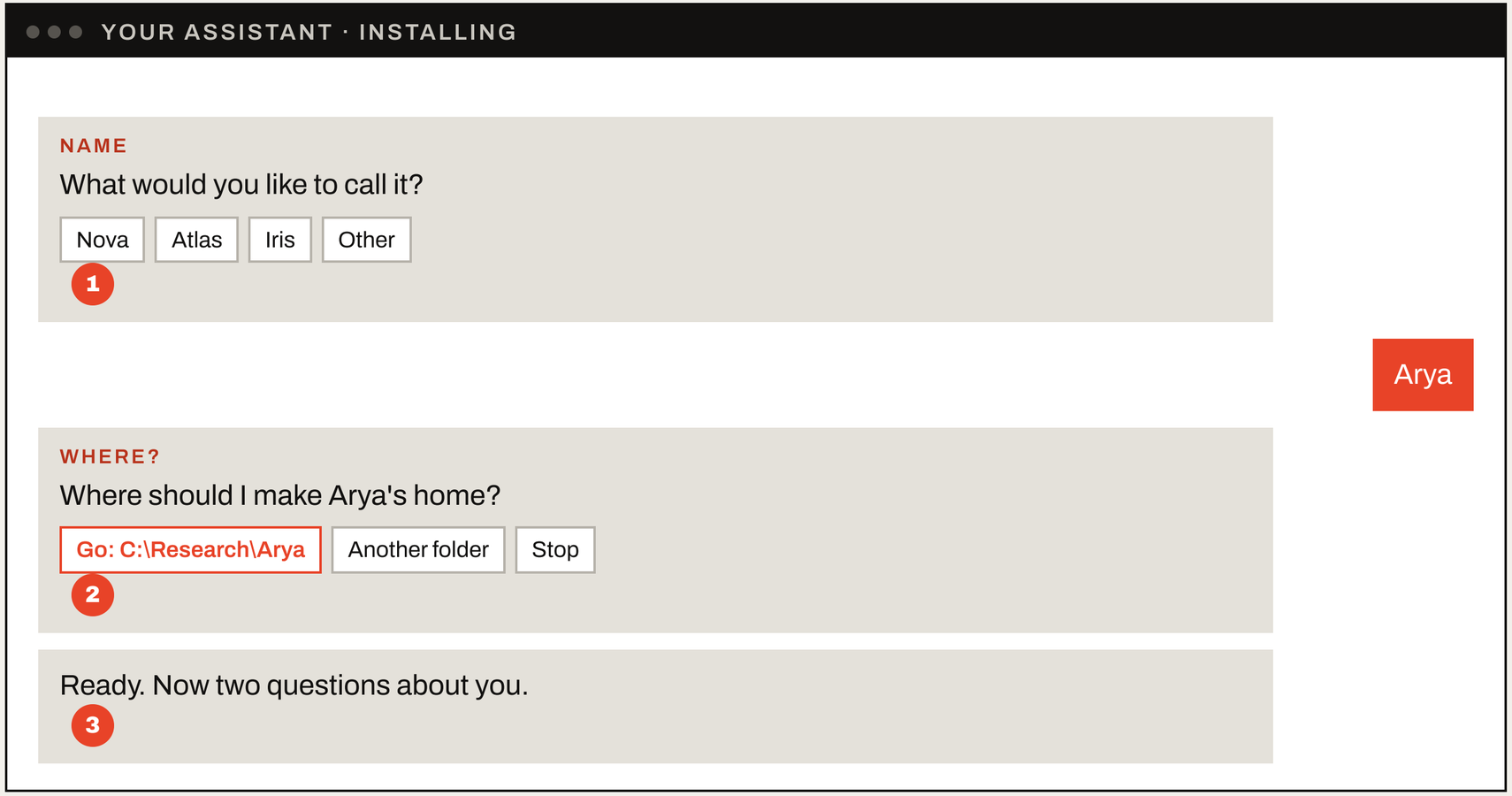
1

1. Copy the README's line; paste it; send.
2. Read each command; allow it.
3. Answer its questions.

github.com/KaxaNuk/KaxaNuk-Researcher · open source · Install

If it stops: the handbook's reference pages.

NAME IT.
CHOOSE WHERE IT LIVES.



1. Name it: an idea, or yours.
2. A short path for its home.
3. Then, two questions.

FIRST YOU, THEN ITS VOICE.

ARYA · SETTING UP

Tell me more about yourself: what you do, and the kinds of projects and topics we'll be working on together.

I'm a portfolio manager: I build strategies in US equity and asset allocation. I want to start a business, with your help. And teach me factor models.

2 OF 2 · VOICE

Brief, push back on evidence

RESEARCHER.md

+ **Works for:** a portfolio manager
+ **Here for:** strategies; a business; learning factor models
+ **Domains:** US equity, asset allocation, entrepreneurship, factor models
+ **Voice:** brief, push back on evidence

A PLACE FOR EVERYTHING.

Made where you said in the interview: on Windows, C:\Research\Arya. Open it and look.

FOLDER	WHAT GOES THERE	WHO WRITES IT
`RESEARCHER.md`	Who you are, its voice, your rules.	It, from the interview; you, by hand.
`Sources/`	What you give it: papers, books, clippings.	You.
`Knowledge/`	Its notes, each linked to a source; INDEX.md, LOG.md.	It, after `read`.
`Studies/`	Decisions weighed, for and against.	It, with `study`.
`Lessons/`	Lessons and their quizzes.	It, with `teach`.
`Philosophy/`	How you think and invest, in your words.	You, with `philosophy`.
`Briefs/ · Portfolio/`	Your daily brief; your holdings' rules. They stay on this computer.	It; you.

THE RULE Sources/ → read → Knowledge/ → ask, learn, decide.

GIVE IT SOMETHING TO READ.

C:\RESEARCH\ARYA\SOURCES

```
Books/
  Markowitz_1959.pdf
Papers/
  Mangram_2013.pdf
  Arnott_…_2019.pdf
Clippings/
```

ARYA · HAND-OVER

FIRST STEP

I have a document RECOMMENDED

1

Or open a new chat in my home.

2

1. Now: attach a PDF here.
2. Or later: a new chat.

Download them: Markowitz (1959), Cowles Monograph 16 · Mangram (2013), SSRN 2147880 · Arnott, Harvey & Markowitz (2019), SSRN 3275654

Three to start

Markowitz, risk weighed with return; Mangram, the same made simple; Arnott, how to test without fooling yourself.

SEVENTY YEARS OF INVESTING IDEAS.

1

1952 MARKOWITZ

RISK WITH RETURN

Weigh risk with return.

Portfolio Selection

1964 SHARPE

A PRICE FOR RISK

Only market risk is paid: beta.

Capital Asset Prices

1968 JENSEN

THE FIRST TEST

Most funds did not beat the market.

The Performance of Mutual Funds

1970 FAMA

THE NULL

Prices reflect what is known.

Efficient Capital Markets

1979 KAHNEMAN & TVERSKY

THE PEOPLE

Losing hurts more than winning feels good.

Prospect Theory

1980 GROSSMAN & STIGLITZ

WHY AN EXCESS RETURN MUST EXIST

Costly information: someone is paid.

On the Impossibility of Informationally Efficient Markets

1993 FAMA & FRENCH

FACTORS

Size and value, beyond beta.

Common Risk Factors

1993 JEGADEESH & TITMAN

MOMENTUM

What rose tends to keep rising, for months.

Returns to Buying Winners and Selling Losers

1997 CARHART

THE FOURTH FACTOR

Fund streaks: factors and costs, not talent.

On Persistence in Mutual Fund Performance

1997 SHLEIFER & VISHNY

WHY IT PERSISTS

Seeing it is not holding it.

The Limits of Arbitrage

2004 LO

THE SYNTHESIS

Strategies crowd, fade, and come back.

The Adaptive Markets Hypothesis

2016 MCLEAN & PONTIFF

DECAY

Smaller once published.

Does Academic Research Destroy Stock Return Predictability?

2

2016 HARVEY, LIU & ZHU

TOO MANY TESTS

Most factors may be false: t above 3.0.

… and the Cross-Section of Expected Returns

3

2023 JENSEN, KELLY & PEDERSEN

THE REBUTTAL

Most do replicate.

Is There a Replication Crisis in Finance?

HOW IT EVOLVED

From risk and return to factors, and today's debate.

THEREFORE

① is your first read; then ② and ③ argue it out.

A PAPER IN, A CITED NOTE OUT.

ARYA · ITS HOME

END OF THE INTERVIEW · FIRST STEP

I have a document

1

read Sources/Papers/Mangram_2013.pdf

2

Which question is it for? Your go?

go

3

Saved.

NEW FILES

```
Sources/Papers/
  Mangram_2013.pdf
Knowledge/Finance/
  Mangram_2013_….md
  concept pages
Knowledge/INDEX.md
Knowledge/LOG.md
```

1. At the interview's end: I have a document.
2. Or: save it in Sources/; say read.
3. Read its plan; give your go.

BEYOND READING: ITS OTHER SKILLS.

Ask it “what can you do?” to see them all.

QUERY

Ask: “why diversify?”

`answers`

TEACH

A lesson on diversification.

`with a quiz`

STUDY

Test this idea?

`for and against`

PHILOSOPHY

How you invest.

`in your words`

BRIEF

Your day and markets.

`every morning`

REMEMBER THIS

A rule of yours.

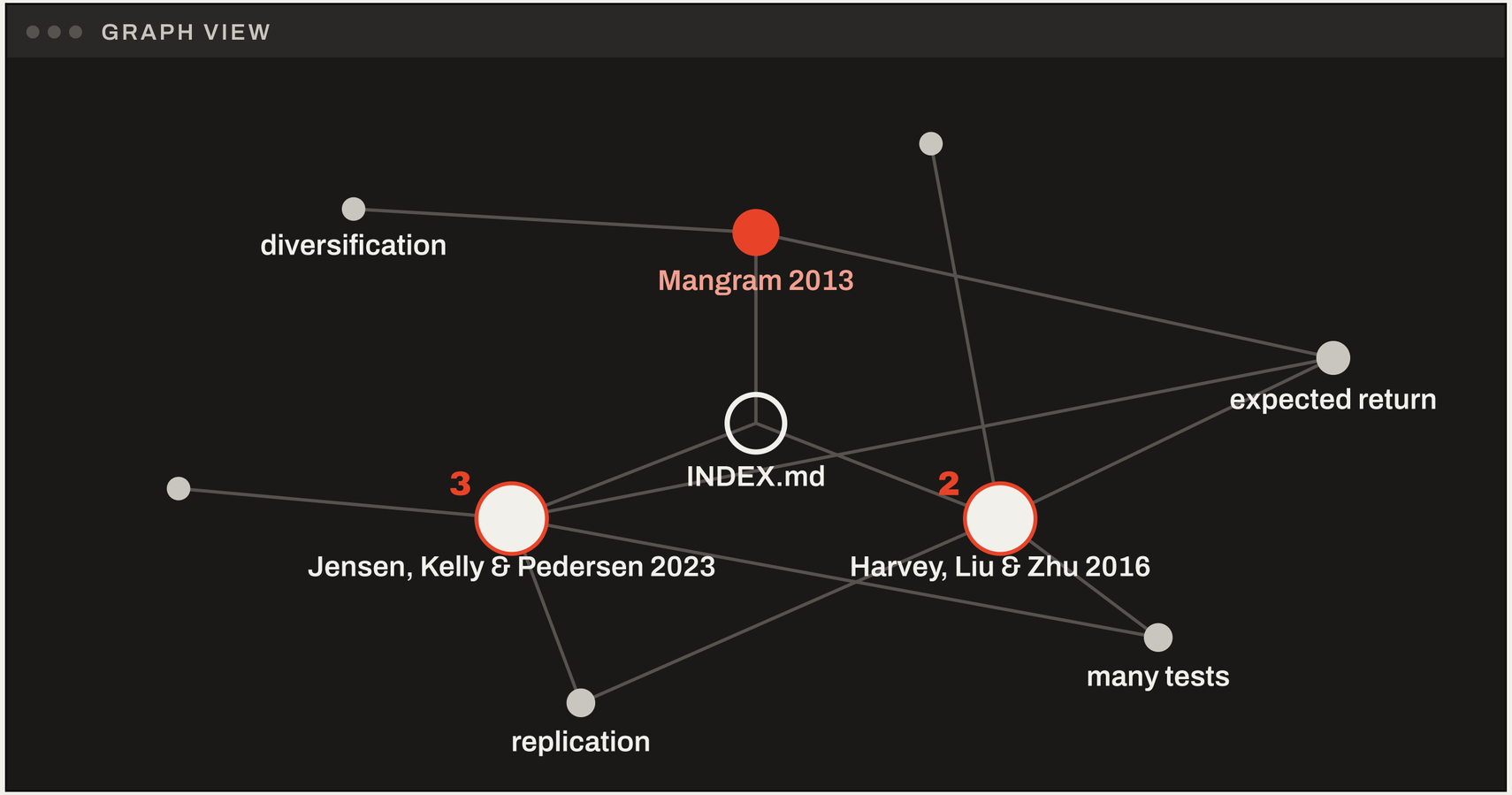
`kept`

YOUR CONTEXT, IN EVERY PROJECT.

	A PLAIN CHAT	YOUR RESEARCHER
Your library	Forgets what it read.	What you read becomes notes on your computer; answers cite them and name what is missing.
You	Starts from zero every chat.	Your questions, your voice and your rules, in your own files.
Every project	One folder, one chat, alone.	The same researcher in every folder: say its name.
Claude, Codex, Gemini	Switch, and start over.	Install it on each: it keeps all of it.
A method	Writes when asked.	A plan and your go first; a saved version of each change.
Day by day	Every chat starts the same.	Its context grows with each use: better results, and yours alone.

IT RUNS ON THEM You still use Claude, Codex or Gemini; your researcher adds what they don't keep from one chat, or one folder, to the next.

WATCH YOUR KNOWLEDGE GROW.



1. Open its home as a vault.
2. Follow links, backlinks, search.
3. Edit RESEARCHER.md by hand.

`obsidian.md` `Graph view` `Backlinks` `Search`

No code needed: Obsidian is an easy way to read and edit RESEARCHER.md and Philosophy/.

TESTING AN IDEA, MADE SIMPLE.

YOUR RESEARCHER · THE INVESTMENT LAB

Testing an investment idea takes many steps and many tools. Your researcher walks each step with you and drives the Lab's tools, so the method stays sound.

IDEAS ARE SIMPLE. TESTING THEM WELL IS NOT.

IDEA	THE RULE	WHAT IT TESTS	ITS CONTROL	BENCHMARK
Bonds or cash	Hold long bonds while the yield curve steepens; else cash.	Does the curve warn ahead of bonds?	Always hold the bonds.	Cash.
USD/MXN	Hold dollars while USD/MXN is above its 200-day average; else pesos.	Does the peso's trend persist?	Always hold pesos.	Pesos.
Stocks: Golden-Flow	The most traded KN US Equity Core stocks, while their trend is up.	Does the trend add to liquidity?	The same, trend ignored.	KN US Equity Core.
Stocks: value with growth	The cheapest by price to earnings, only if their earnings grow; rebalanced each quarter.	Does growth tell cheap from broken?	The cheapest, without the growth filter.	The whole market.
Portfolio: 60/40	60% stocks and 40% bonds, rebalanced each year.	Does diversifying cut risk without costing return?	The same, never rebalanced.	Stocks alone.

THE LAB MAKES IT SIMPLE a backtest without the common mistakes, and an attribution that separates what your rule adds from factor exposure. Ideas and hypotheses, validated with evidence, not beliefs.

SIX LIBRARIES, FROM DATA TO ATTRIBUTION.

CURATOR	REFINERY	ANALYZER	CONSTRUCTION	BACKTEST	ATTRIBUTION
DATA CURATOR Retrieves, validates and unifies data from several providers in one scheme; add your own calculations. **Acquire. Validate. Unify.**	**DATA REFINERY** Enriches that data with sectors, macro or alternative data, and builds features across assets and time. **Enrich. Combine. Scale.**	**DATA ANALYZER** Explores your data before your strategies use it: outliers, stability and patterns. **Explore. Detect. Understand.**	**PORTFOLIO CONSTRUCTION** Turns your signals into portfolios: universe, selection, weights, constraints and rebalancing. **Select. Weight. Rebalance.**	**BACKTEST ENGINE** Simulates those portfolios with execution prices, commissions and slippage; 30+ metrics against your benchmark. **Simulate. Measure. Validate.**	**ATTRIBUTION ANALYSIS** Explains where the return came from: allocation and selection, and each factor against what is stock-specific. **Decompose. Explain. Improve.**

Our building blocks

all modular: use every library, or only the ones you need.

ONE LAB, WITH OR WITHOUT CODE.

INVESTMENT LAB

UNIVERSE
Creator

DATA
Curator
Refinery
Analyzer

PORTFOLIO
Construction

BACKTEST
Parameters

ATTRIBUTION
Attribution

WHAT YOU BUILD THERE

WITHOUT CODE
Build strategies and experiments step by step, with clicks.

WITH CODE
Debug every step, and customise any idea the platform can't build.

ONE ENGINE
The same libraries under both.

Pick your way in
by what you need to build: the result is the same.

ONE TO STUDY. ONE TO BUILD.

INIT-EXAMPLE · GOLDEN-FLOW

```
Bibliotheca/
OBJECTIVE.md  written
Universe/  built
Data/  curated
  Curator/
  analyzer.ipynb
Experiments/  run
  BLUEPRINT_1.md
  FINDINGS_1.md
RESULTS.md  measured  1
```

INIT-STRATEGY · YOURS

```
Bibliotheca/
OBJECTIVE.md
Universe/
Data/
  Curator/
  analyzer.ipynb
Experiments/
  BLUEPRINT_1.md
  FINDINGS_1.md
RESULTS.md  2
```

1. Study it: read it, run it, never build on it.
2. Build yours: the same folders, empty.

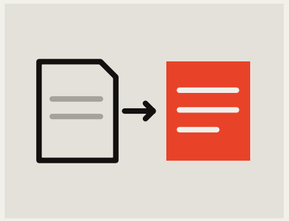
Open both

the example beside yours, one skill for each step.

EIGHT STEPS: KAXANUK'S PROCESS.

YOUR RESEARCHER | THE LAB, OR YOUR CODE | THE GATE | OUTSIDE THE COURSE

1 — 2 — 3 — 4 — 5 — 6 — 7 — 8

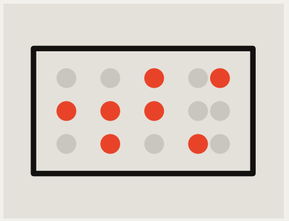


RESEARCH

S02

Your objective, from your library · OBJECTIVE.md

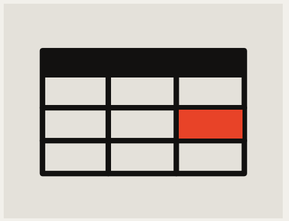
S02 STARTS HERE



UNIVERSE

S02

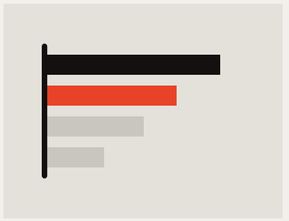
Which companies, when; delisted kept · Universe/



DATA

S03

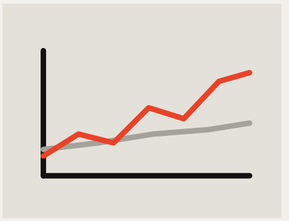
Your variables, and whether they carry signal · Data/



PORTFOLIO

S04

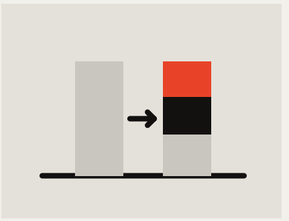
Your hypothesis, then portfolios · Experiments/



BACKTEST

S05

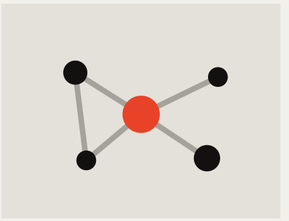
Your portfolio through time, net of costs



ATTRIBUTION

S05

Where the return came from



PAPER TRADING

S06

Tested without real money · Paper_Trading/



PRODUCTION

outside the course

The sessions ahead

S02 to S06 walk these steps, one by one, on your own strategy.

ARYA WALKS YOU STEP BY STEP.

YOUR EDITOR · GOLDEN-FLOW

```
Bibliotheca/
Universe/
Data/
Experiments/
   Experiment_1/
      BLUEPRINT_1.md
      JOURNAL_1.md
      FINDINGS_1.md
      experiment_1.ipynb
Paper_Trading/
Config/.env.template
```

Arya, where am I?

Step 5, the backtest: your portfolios are ready. 1

Plan: one call of `backtest-engine-runs`. Go? 2

go

Done. Checked: delisted names kept, costs in. 3

1. Knows your step.
2. Knows how to use each library.
3. Guards the common mistakes.

IT BEAT ITS CONTROL, MOSTLY WITH THE MARKET.

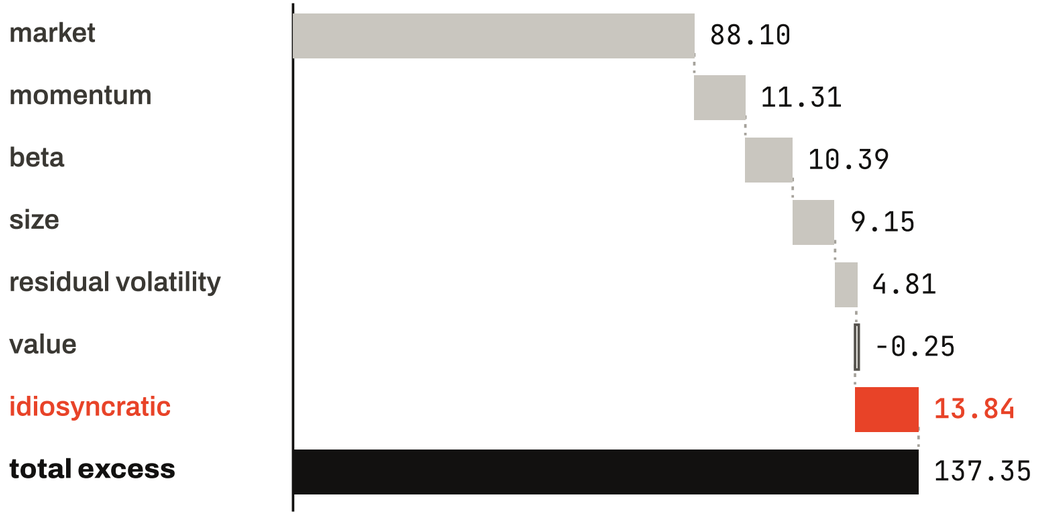
GOLDEN-FLOW · EXPERIMENT 1

Golden-Flow · FINDINGS_1.md

SHARPE · 2015-01-02 TO 2026-06-01



THE RULE'S EXCESS RETURN, BY FACTOR · POINTS



CAGR	MAX DRAWDOWN
20.33%	-32.47%

Hypothetical: a backtest, net of costs, in sample. Points: daily returns summed, not compounded.

BACK UP YOUR RESEARCHER AND YOUR STRATEGIES.

YOUR RESEARCHER · ITS HOME

- **A private copy on GitHub** backup
- read: Mangram EXAMPLE
- The interview EXAMPLE
- First version EXAMPLE

YOUR STRATEGY · ITS REPOSITORY

- **A private copy on GitHub**
- Experiment 1: findings EXAMPLE
- Experiment 1: blueprint EXAMPLE
- First version EXAMPLE

Why

a new computer, a mistake undone, your whole history.

THE SESSION'S IDEAS, IN YOUR WORDS.

01
What makes your researcher yours?

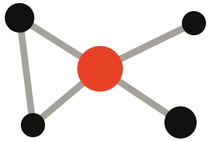
02
How do you train it?

03
What does the Lab do, step by step?

04
Where does each part of a strategy live?

Next, S02: your objective and your universe.

BEFORE SESSION 02: ARRIVE READY.



01 TRAIN IT

- Read the debate: Harvey, Liu & Zhu (2016), then Jensen, Kelly & Pedersen (2023).
- Ask for `teach` on each.



02 ANY PROJECT

- Open it in any project and say its name.
- Not investing yet? It leads you there.



03 READY FOR S02

- Your idea in one line: what it buys, and why.
- The benchmark it must beat, and why.
- `init-example`: browse Golden-Flow.
- First Steps: CF P03, MF P01, FM P06.

YOU HAVE YOUR RESEARCHER. NOW TEST YOUR IDEAS.

Train it every day, take it to every project, and validate your ideas with evidence.

LEARN

The Bootcamp continues, then the pillars.

`github.com/KaxaNuk/Investment-Bootcamp`

TEST

Your strategy on the Lab; universes and factors from Analytics.

`kaxanuk.mx/lab` · `kaxanuk.mx/analytics`

TELL US

What your researcher helped you do.

QUESTIONS `workshops@kaxanuk.mx`

NEXT

Session 02 · Your Strategy's Objective and the Investable Universe

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Nothing here is a recommendation to buy, sell or hold any security; the ideas shown are ideas to test, backtests, Golden-Flow's included, are hypothetical, in sample and never traded, and illustrative charts carry no data. Your researcher can be wrong; every decision is yours. What your researcher reads goes to the company behind your assistant, never to KaxaNuk.

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